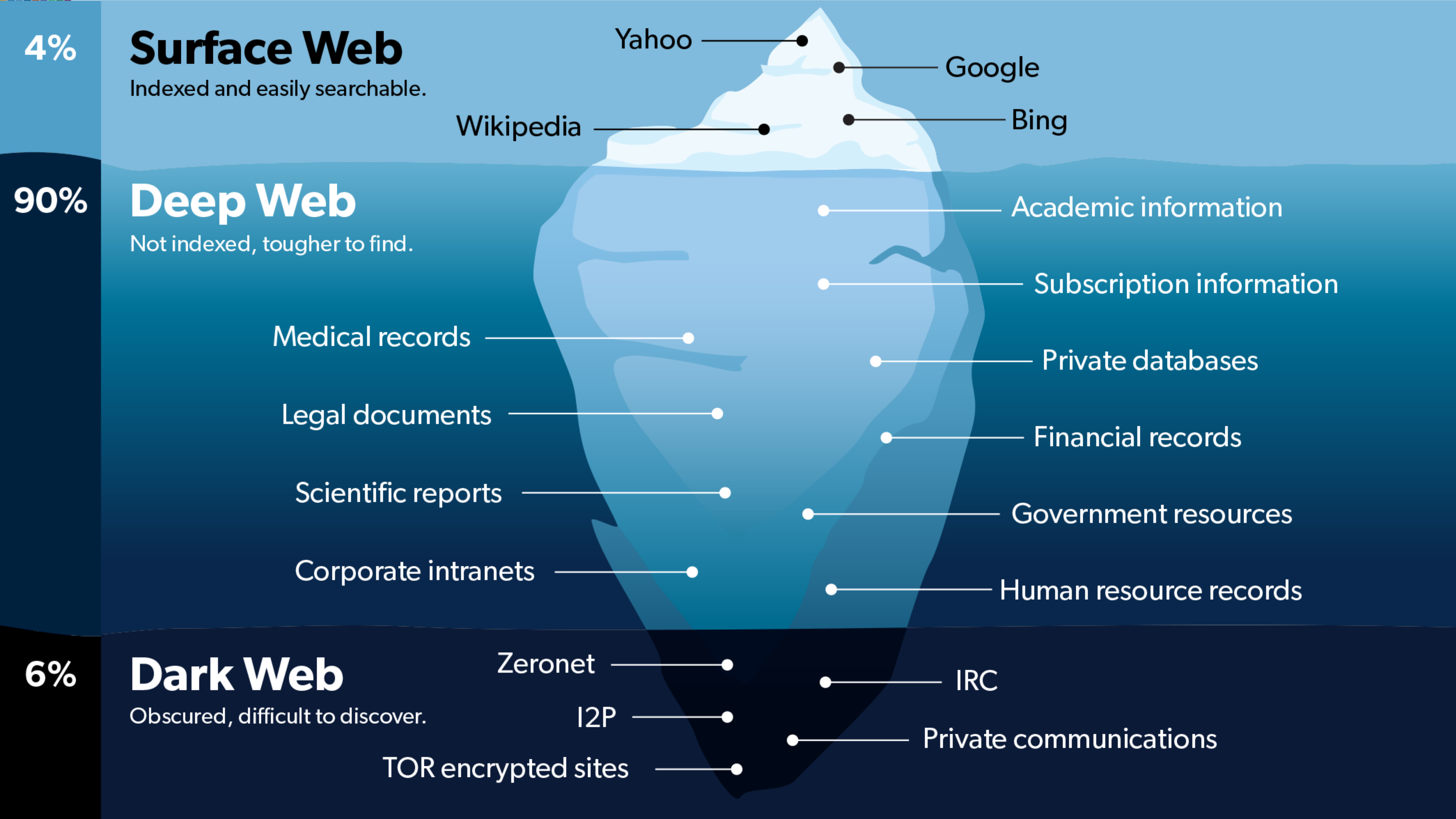
UNIVERSITÀ DEGLI STUDI DI MILANO

Tor Project

Onion Routing Overview

Dark and Deep Web

- Dark/deep web is a part of the Internet (websites, resources, documents) that is not accessible through standard browsers and search engines
- Usually linked to illegal activities, censorship, private and anonymous communications


4%
Surface Web
Indexed and easily searchable.
Yahoo
Google
Wikipedia
Bing
90%
Deep Web
Not indexed, tougher to find.
Academic information
Subscription information
Medical records
Private databases
Legal documents
Financial records
Scientific reports
Government resources
Corporate intranets
Human resource records
6%
Dark Web
Obscured, difficult to discover.
Zeronet
IRC
I2P
Private communications
TOR encrypted sites

.onion

- The dark web is composed of a series of hidden websites, forums, illegal markets, and other services not accessible through conventional browsers
- We can access these resources via **.onion** websites, which are supported by the Tor browser (a modified version of Firefox)
- Unlike the conventional Web, there are **no central authorities** (ISPs) regulating the Dark Web; it has a decentralized structure, making it difficult to monitor

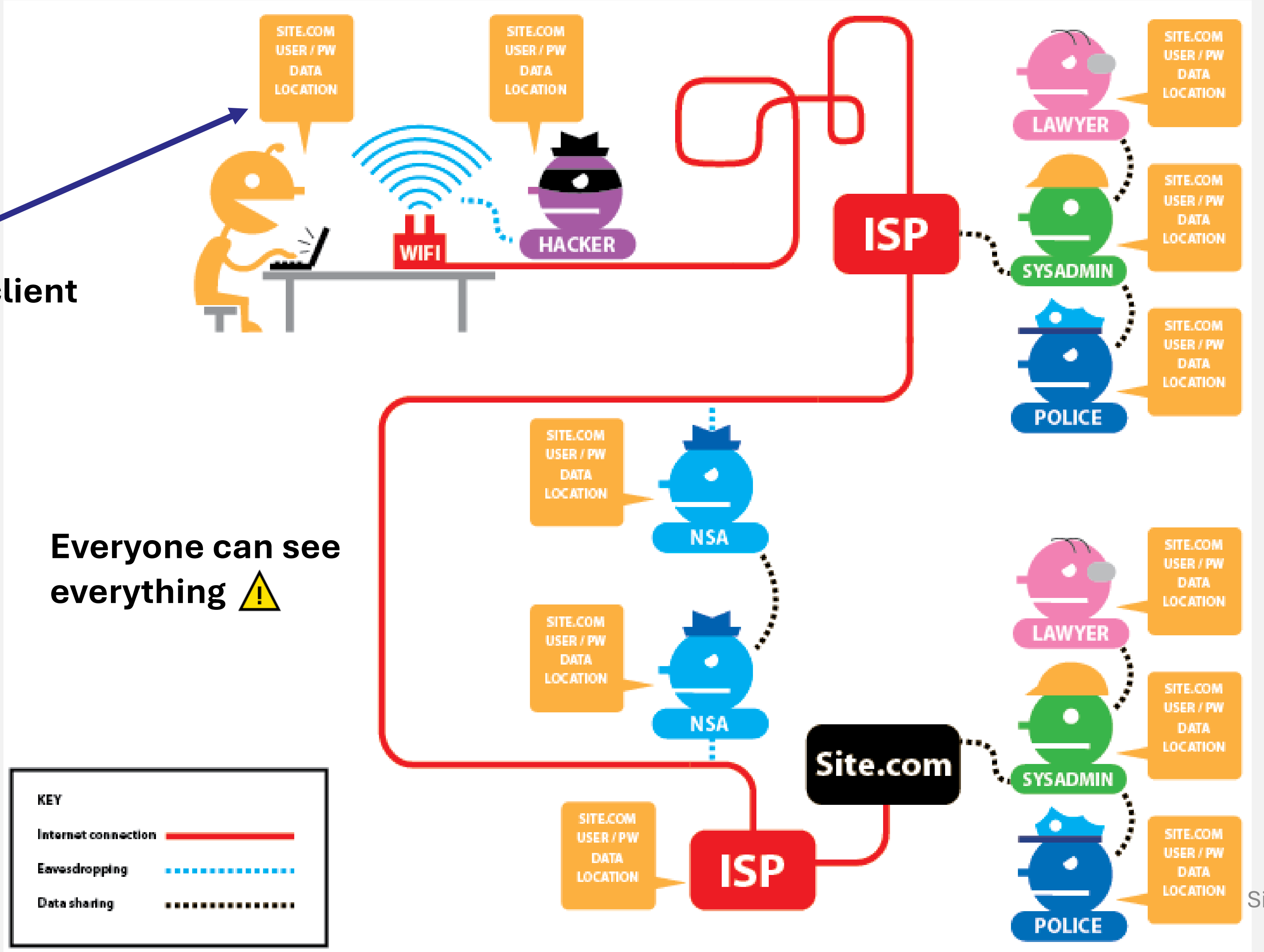
Tor, the onion routing

- Tor (The Onion Routing) is an implementation of the onion routing protocol developed in the mid-90s by the US Naval Research
- Idea: **make a message bounce between different nodes** to make it very difficult to understand who the communication interlocutors are (i.e an IP address)
- Unlike an encrypted message, which can be intercepted by anyone but readable only by the recipient, the TOR protocol also aims to avoid identifying the interlocutors in the conversation.

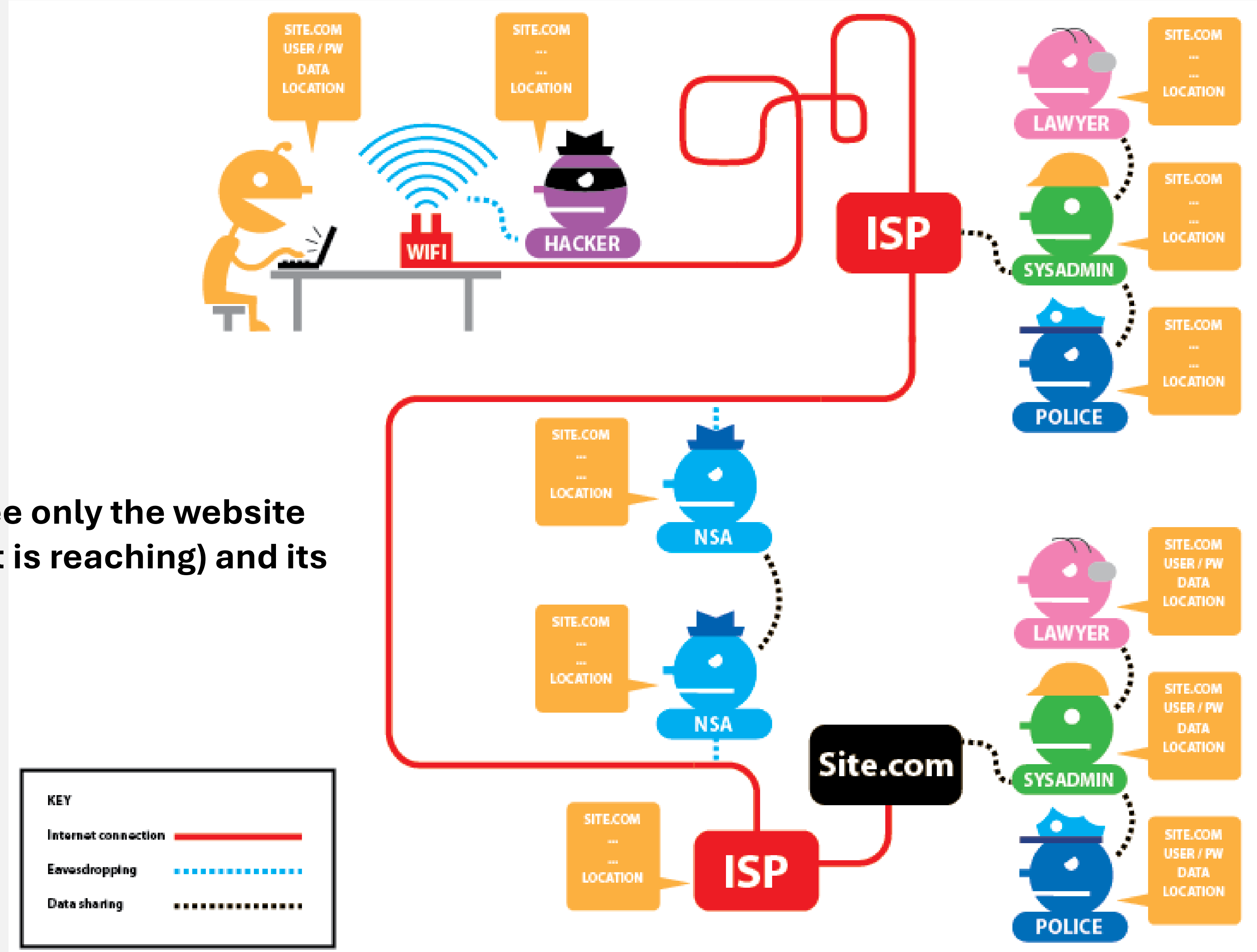
An internet connection without HTTPS

Location is the client IP address

Everyone can see everything ⚠



An internet connection with HTTPS



Anyone can see only the website (that the client is reaching) and its IP ADDRESS

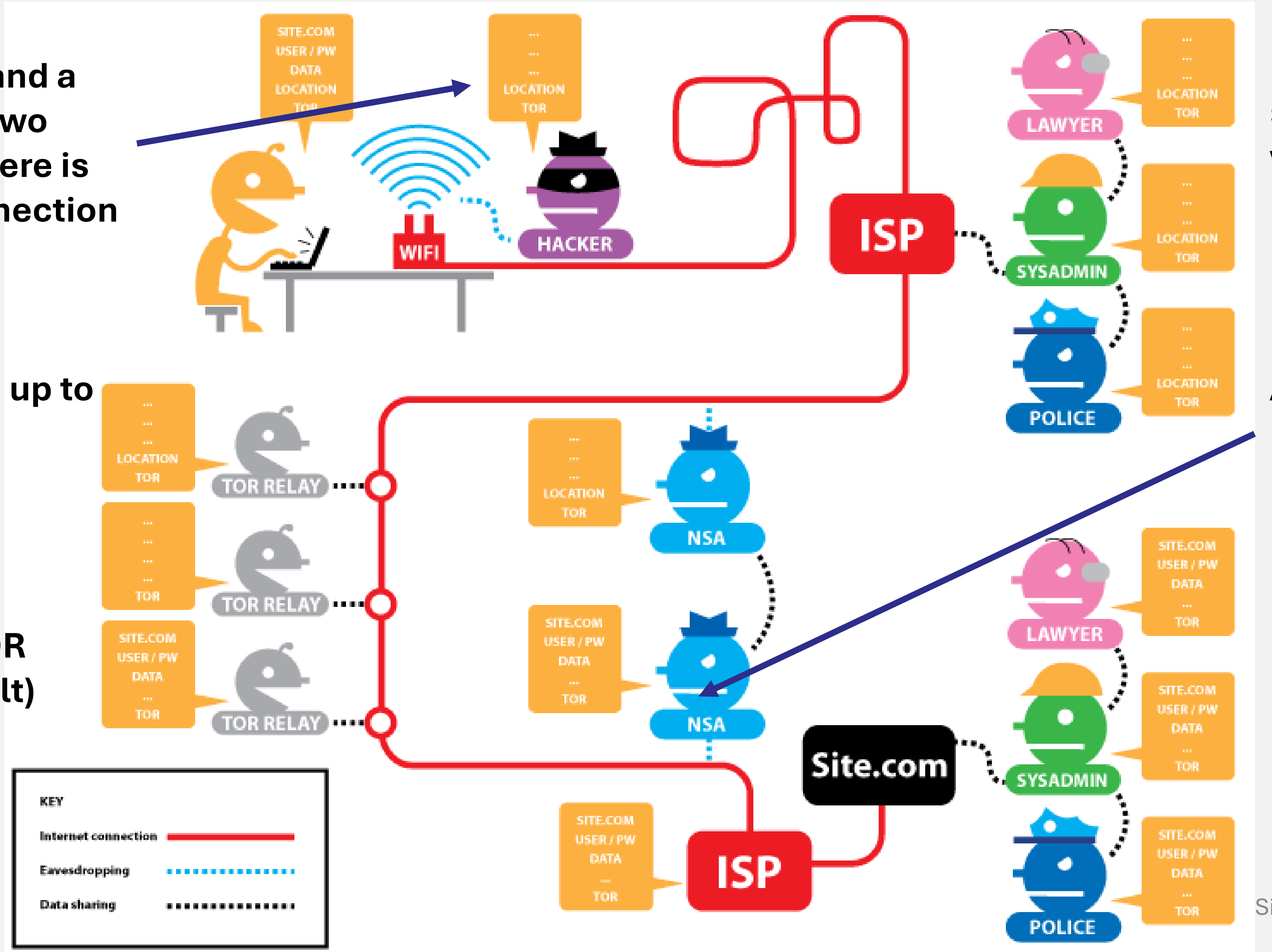
The website host can see everything and share that information to third parties

An internet connection with TOR and without HTTPS

Between a client and a relay or between two adjacent relays there is an encrypted connection

Location is known up to the first relay

Introduction of TOR Relays (3 by default)



Intermediate relays cannot spoof the website

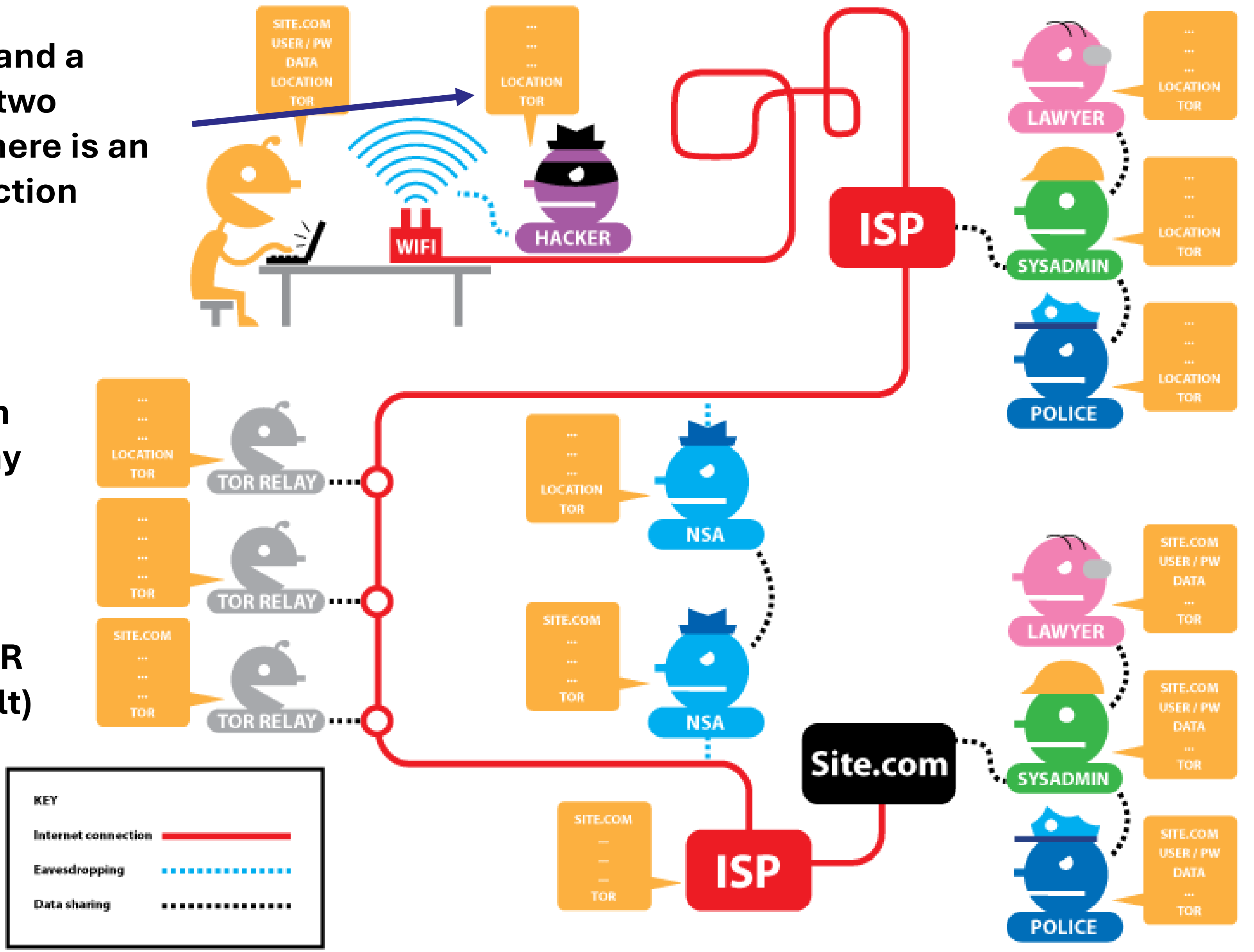
After the last relay, data is no more encrypted

An internet connection with TOR and with HTTPS

Between a client and a relay or between two adjacent relays there is an encrypted connection

Location is known up to the first relay

Introduction of TOR Relays (3 by default)



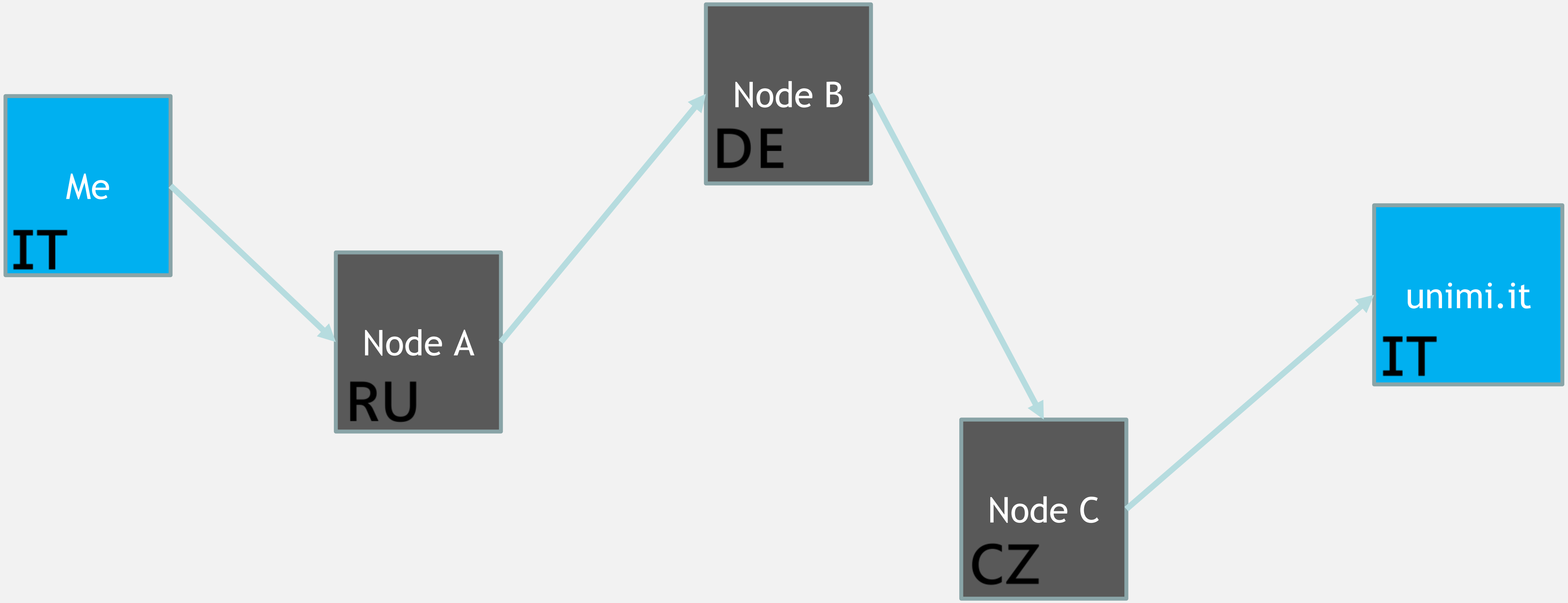
Data is visible only by the host and possible third parties

Basic introduction

- Relay (or onion router) : a server that joins the TOR network
- Client: a browser that leverages the TOR network to access a website

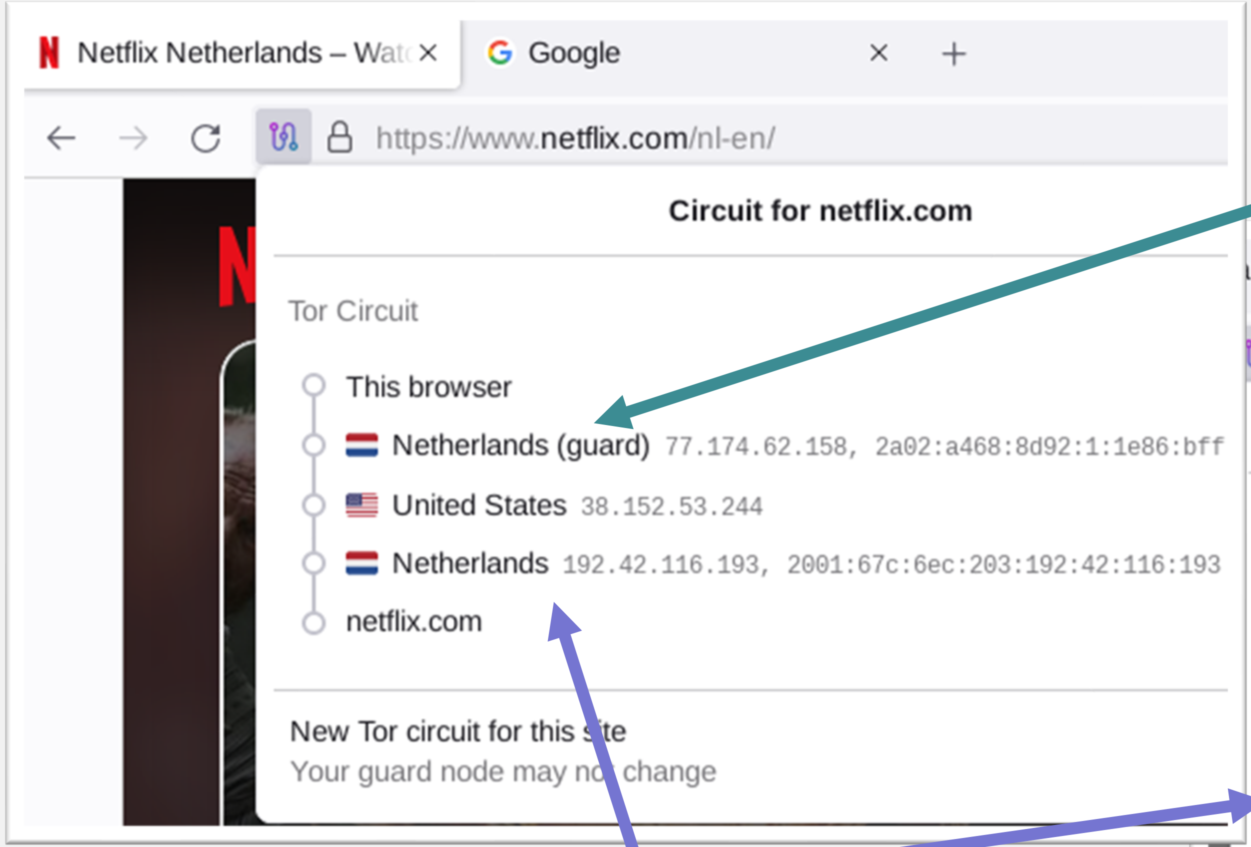
- To anonymize its traffic, a client needs to create a **multi-hop cryptographic** structure called "**circuit**", typically composed of 3 relays: messages bounce around these nodes until they reach the recipient
- **A relay knows its predecessor and successor in the circuit, but not the others**

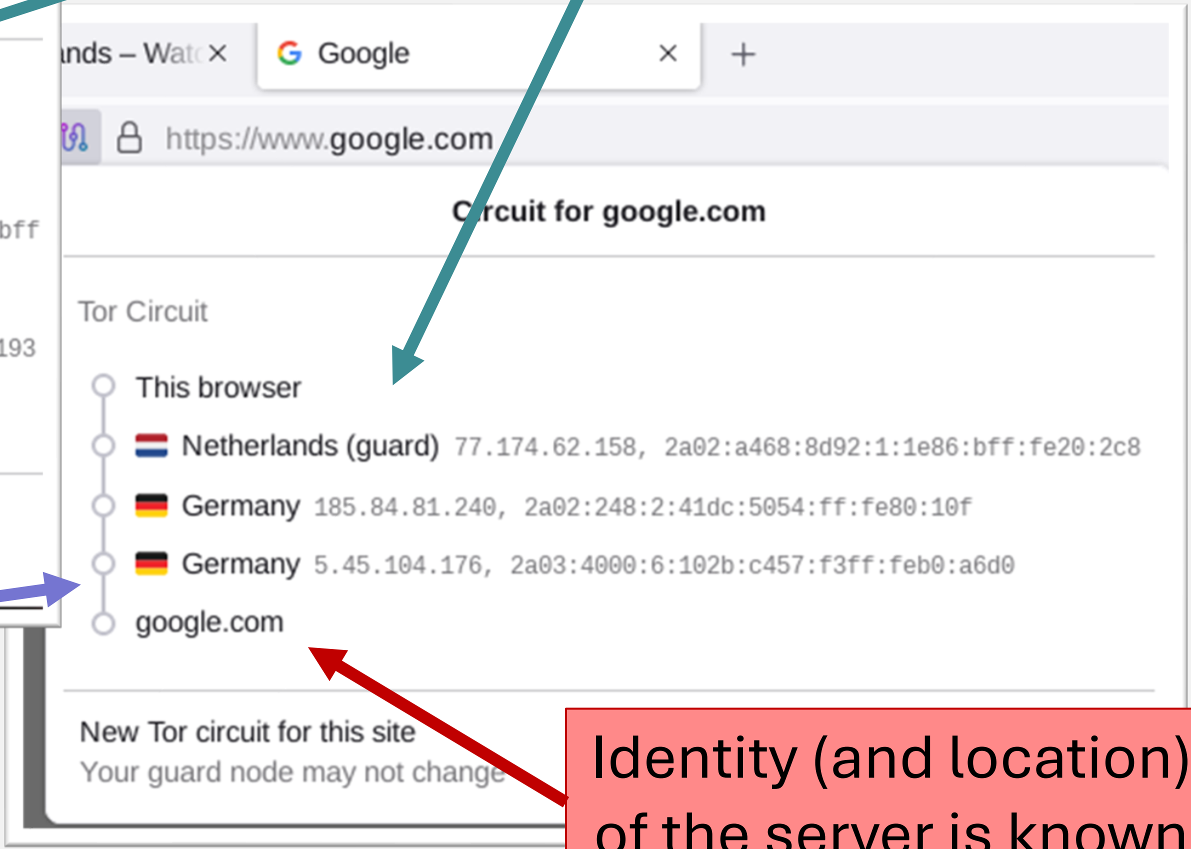
Tor, the onion routing



The circuit

Only the guard knows the “identity” of the client





Only the last relay knows the destination

Identity (and location) of the server is known

Creation of a circuit composed of N relays

- Choose an onion router as the end node R_n (from a known list)
- Choose a chain of $N - 1$ routers ($R_1 \ldots R_{n-1}$)
  - The client chooses the relays, so he knows the public keys of each relay
- Send a **creation message** to the first router R_1
- For each subsequent router $R_2 \ldots R_n$, extend the circuit
  - Create a specific message for R_m and encrypt it with R_m's public key
  - Send the message along the existing circuit

```
Client sends relay cell:
   For I=N...1, where N is the destination node:
      Encrypt with Kf_I.
   Transmit the encrypted cell to node 1.
```

Creation of a circuit composed of N relays

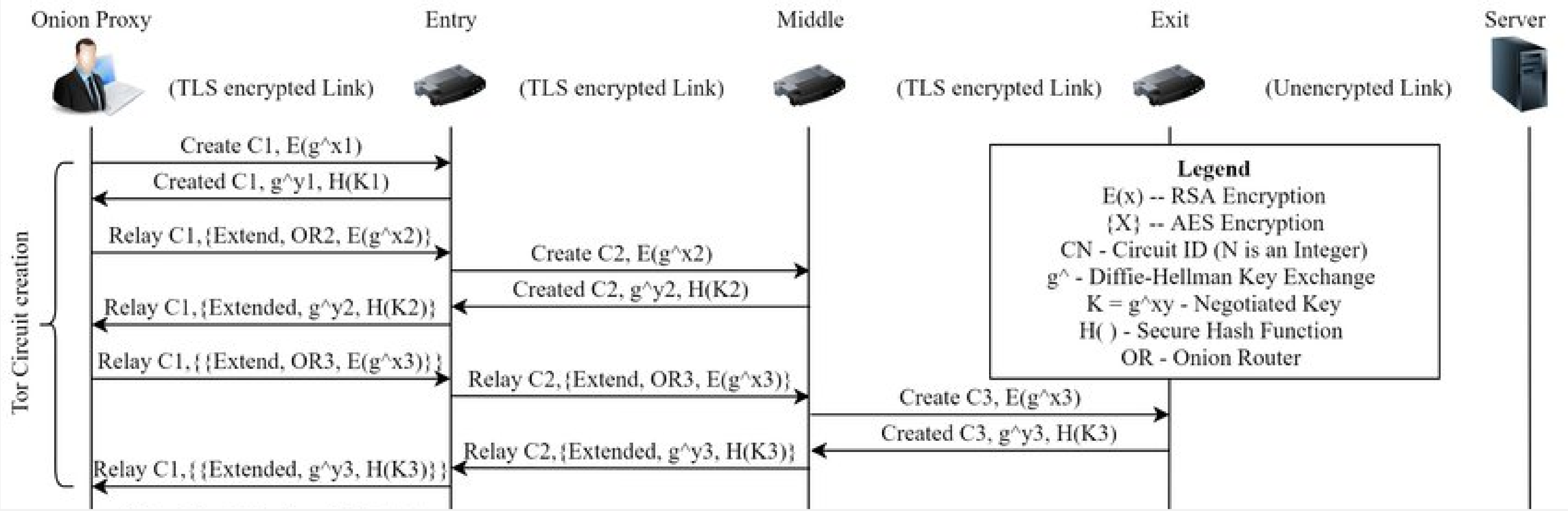
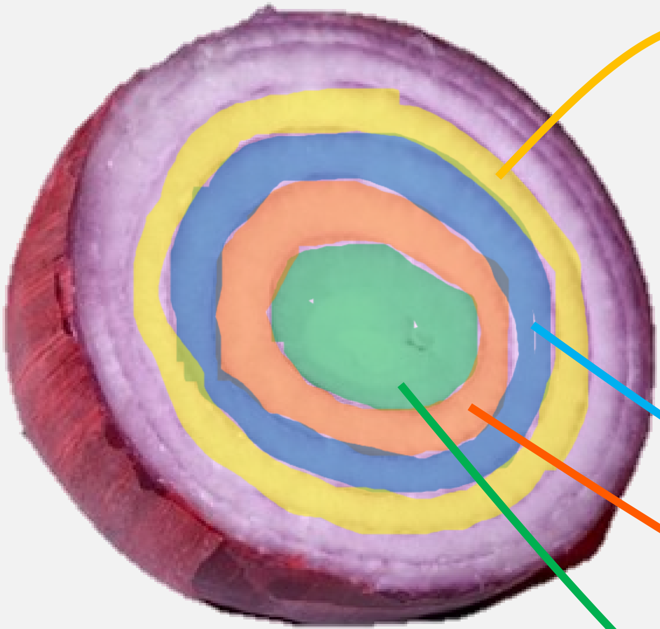


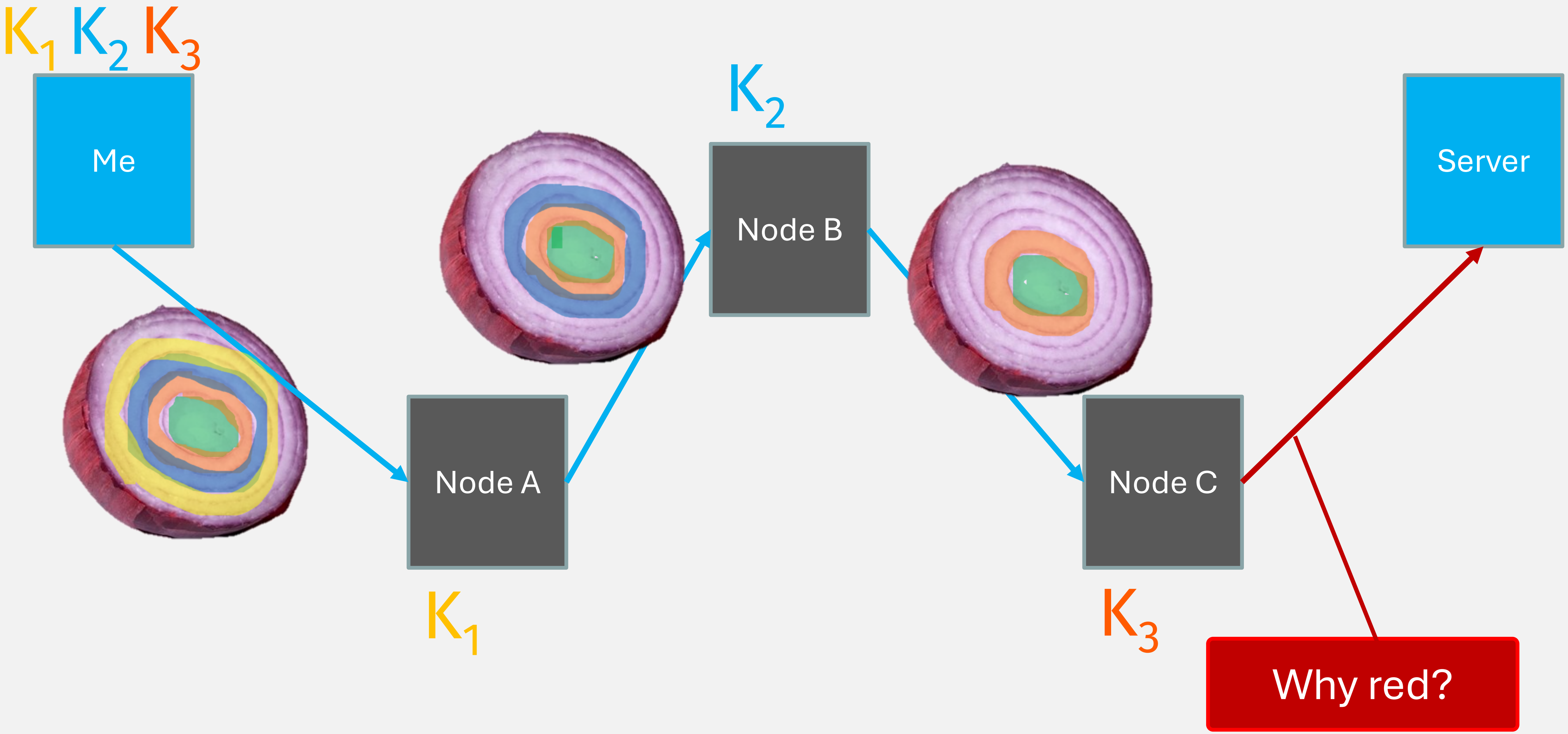
Image taken from: : I. Karunanayake, N. Ahmed, R. Malaney, R. Islam and S. K. Jha, "De-Anonymisation Attacks on Tor: A Survey," in IEEE Communications Surveys & Tutorials

Tor, the onion routing



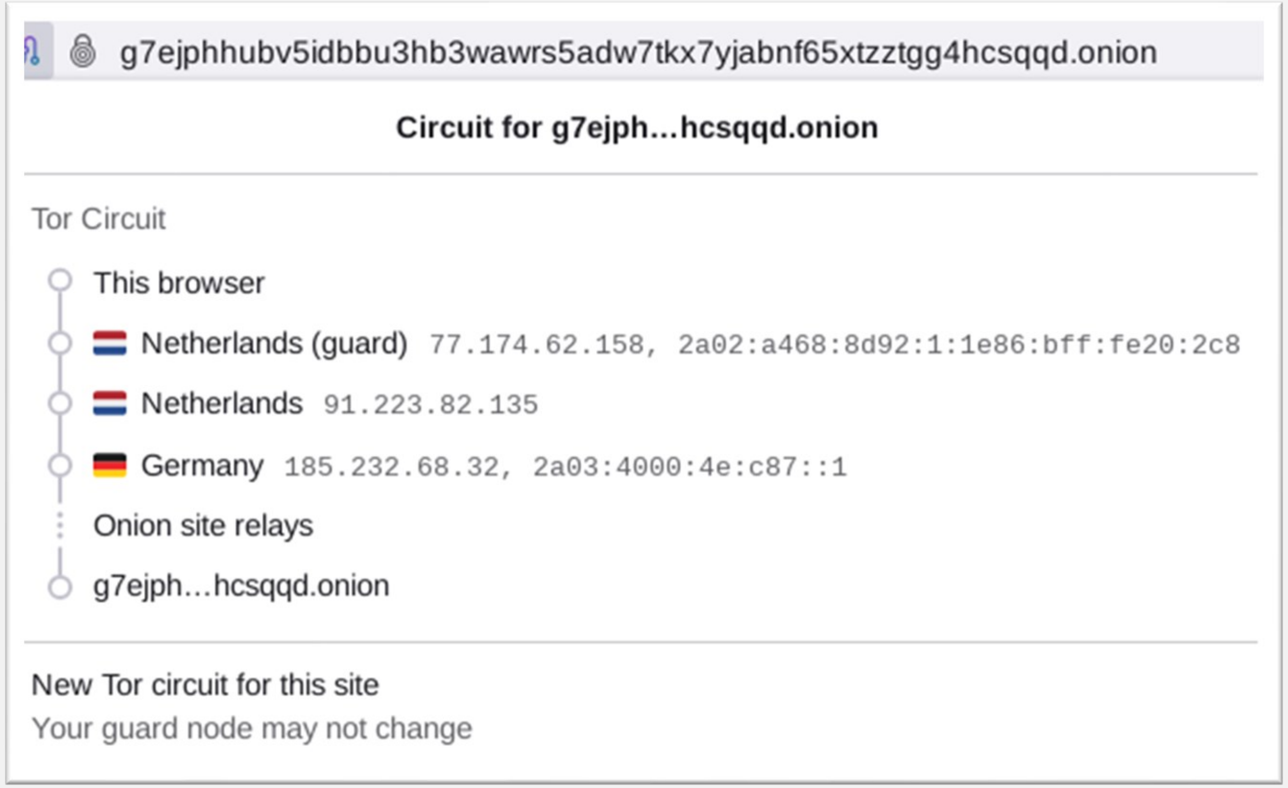
$$E_{k1}(E_{k2}(E_{k3}(Message)))$$

Tor, the onion routing

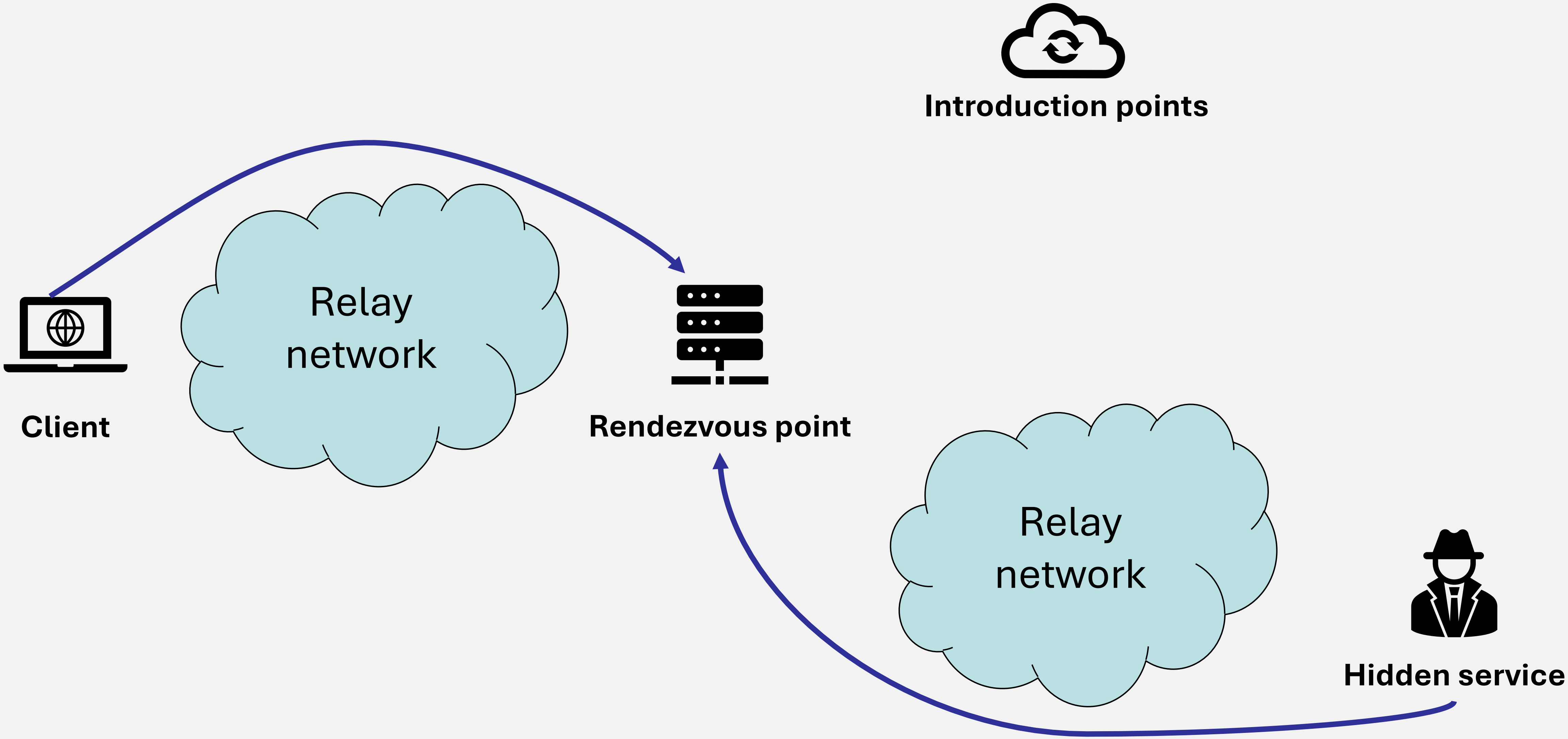


Onion services

- What if also the server wants to hide his identity and its location?
- Hidden Services/Tor Rendezvous aim to provide bidirectional anonymity



Hidden service



Hidden service

- An onion service exposes itself by selecting **introduction points**, building circuits to those points, and directing them to pass along introduction requests through those circuits
- To share the introduction points required for establishing a connection, the host publishes a collection of **hidden service descriptors** through dedicated directory nodes
- The client sends an **introduction request** (that includes the target rendezvous point) to the host via an introduction point
- When the host receives an introduction request, it connects to the rendezvous point, allowing the handshake to proceed

Drawbacks

- The more hops we have, the slower the circuit become

  From Tor manual: «Every node is serving a lots of people at once, anyone can join nodes pool with personal computer but Tor nodes should have large bandwidth […] A node, especially user hosted ones, can leaving the circuits at anytime»

- A single nodes might serve a lot of users: need of high bandwidth
- Custom Firefox edition must be installed
- **Not immune to deanonymization attacks**

References

- [A short introduction to Tor - Tor Specifications]
- [Tor Rendezvous Specification - View from 10,000 feet]
- [How HTTPS and Tor Work Together to Protect Your Anonymity and Privacy]